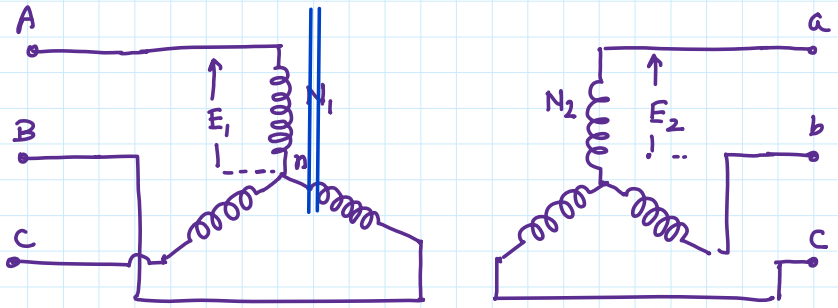
Lecture 41: Induction Machine-Circuit Representation

Induction Machine as a Transformer



$$E_1 = 4.44 f_1 N_1 \phi_{max}$$
$$E_2 = 4.44 f_2 \cdot N_2' \phi_{max}$$

Primary

$N_1' = K_{w1} * N_1$ (Secondary)

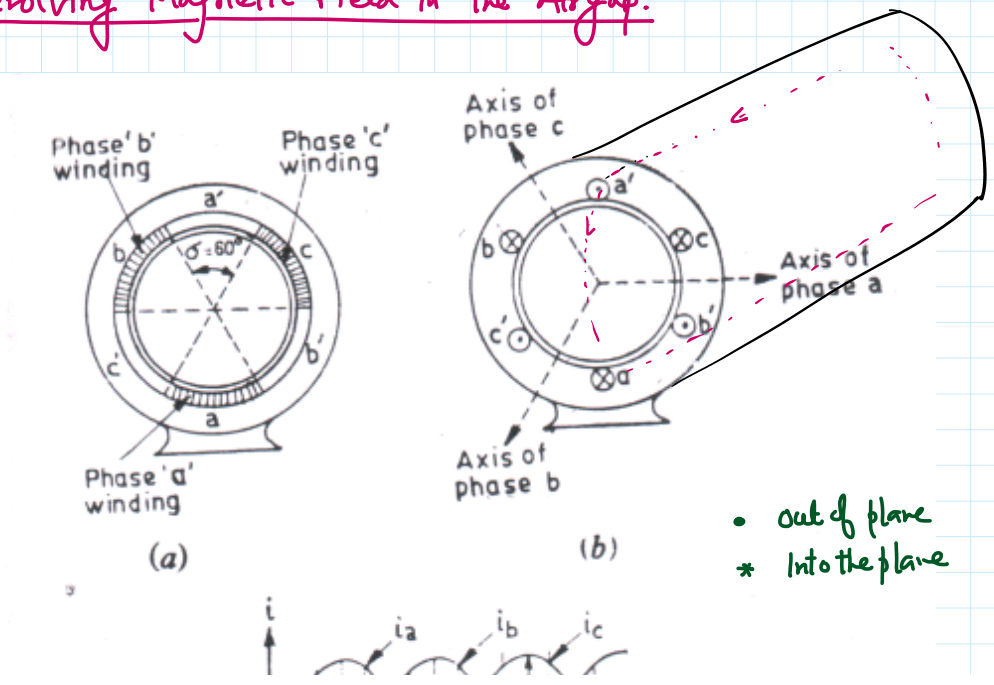
$N_2' = K_{w2} * N_2$

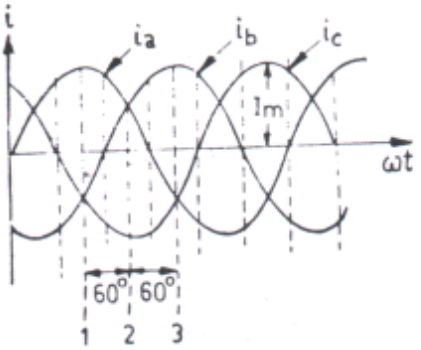
ϕ_{max} = Maximum flux in the core

$$\Rightarrow \boxed{\frac{E_1}{E_2} = \frac{N_1'}{N_2'}}$$

The only difference in transformer and induction machine has rotating part. There is airgap between the rotor and stator, which has more reluctance or more leakage of flux, hence I_0 = No load component of current is more $\simeq$ 30-50% of I rated against 2-6% for transformer.

Resolving Magnetic Field in the Airgap:





Combining Faraday's & Lenz's law: $e = -N\frac{d\phi}{dt} = -L\frac{di}{dt}$

Assuming linearity between ϕ, i, $i = \frac{N}{L}\phi$

Ampere-turns, $Ni = \left(\frac{N^2}{L}\right)\phi$

$MMF = Ni = \frac{N^2\phi}{L}$

$MMF \propto i \propto \phi$

Thus if we can find out combined MMF due to 3-phase winding, it shall also reflect the combined flux or magnetic field.

Let us find out combined MMF (Magnetomotive force).

$i_a = I_m \sin \omega t$
$i_b = I_m \sin(\omega t - 120°)$
$i_c = I_m \sin(\omega t + 120°)$

$F_a = N_{aph} i_a = N_{ph} I_m \sin \omega t = F_m \sin \omega t$
$F_b = N_{bph} i_b = N_{ph} I_m \sin(\omega t - 120°) = F_m \sin(\omega t - 120°)$
$F_c = N_{cph} i_c = N_{ph} I_m \sin(\omega t - 120°) = F_m \sin(\omega t + 120°)$

Now the MMFs, i.e. F_a, F_b, F_c are space distributed sinusoidally. Let α be an angle from the axis of winding of phase-a. The F_a is maximum at $\alpha = 0$ and minimum (negative) at $\alpha = 180°$. Thus

$F_a(\alpha, t) = F_m \sin \omega t \cdot \cos \alpha$
$F_b(\alpha, t) = F_m \sin(\omega t - 120°) \cos(\alpha - 120°)$
$F_c(\alpha, t) = F_m \sin(\omega t + 120°) \cos(\alpha + 120°)$

The resultant flux $F(\alpha, t) = F_a(\alpha, t) + F_b(\alpha, t) + F_c(\alpha, t)$

$= F_m [\sin \omega t \cos \alpha + \sin(\omega t - 120°) * \cos(\alpha - 120°) + \sin(\omega t + 120°) \cos(\alpha + 120°)]$

$F_R(\alpha, t) = \frac{F_m}{2} [2 \sin \omega t \cdot \cos \alpha + 2 \sin(\omega t - 120°) \cdot \cos(\alpha - 120°) + 2 \sin(\omega t + 120°) * \cos(\alpha + 120°)]$

$2 \sin A \cos B = \sin(A - B) + \sin(A + B)$

$= \frac{F_m}{2} [\sin(\omega t - \alpha) + \sin(\omega t + \alpha) + \sin(\omega t - \alpha) + \sin(\omega t + \alpha - 240°)$

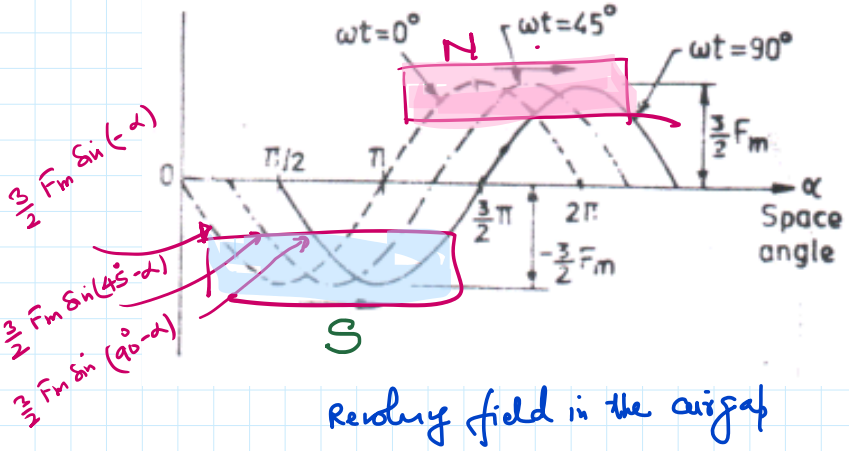
$$= \frac{Fm}{2}\Big[\sin(\omega t-\alpha) + \sin(\omega t+\alpha) + \sin(\omega t-\alpha) + \sin(\omega t+\alpha-240^\circ) + \sin(\omega t-\alpha) + \sin(\omega t+\alpha+240^\circ)\Big]$$

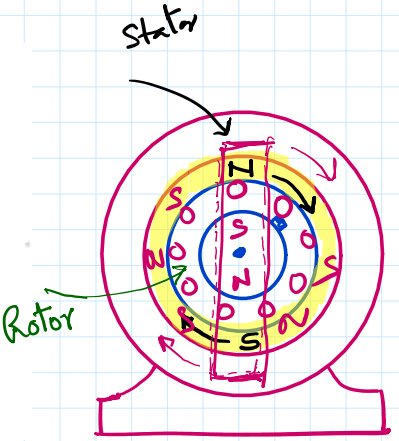
$$\boxed{F_R(\alpha,t) = \frac{3\,Fm}{2}\sin(\omega t-\alpha)}$$

We can see at $\omega t = 0$, $F_R(\alpha,0) = \frac{3}{2} Fm \sin(-\alpha)$

$\omega t = 45^\circ$, $F_R(\alpha,t_1) = \frac{3}{2} Fm \sin(45^\circ-\alpha)$

$\omega t = 90^\circ$, $F_R(\alpha,t_2) = \frac{3}{2} Fm \sin(90^\circ-\alpha)$





Revolving field in the airgap

shortcuts://run-shortcut?name=Revolving%20field

Synchronous Speed (n_s): The revolving field rotates at speed called synchronous speed, ns. It is defined as

$$n_s = \frac{f}{(P/2)} \text{ RPS} \quad \text{or} \quad \frac{2\times 60 f}{P} \text{ RPM} = \frac{120 f}{P} \text{ RPM}$$

P → Total No Poles

where f = Supply frequency of stator voltage or current

Relationship between Electrical and Mechanical speed:

$$\theta_{mech} = \frac{\theta_{elect}}{(P/2)}$$

OR $\theta_{elect} = \frac{P}{2} * \theta_{mech}$

OR $\theta_{elect} = \frac{P}{2} * \theta_{mech}$

OR $\omega_{elect} = \frac{P}{2} * \omega_{mech}$

$\downarrow$

$f_{elect} = \frac{P}{2} * f_{mech}$

$f = \frac{P}{2} * n_s$

$\Rightarrow n_s = \frac{f}{P/2}$ RPS $= \frac{120f}{P}$ RPM

$$n_s = \frac{120f}{P} \text{ RPM}$$

Slip of Induction Motor:

When revolving field revolves at synchronous speed (n_s), it cuts the rotor conductors. Thus induces emf, and due to closed circuit of rotor winding, current starts flowing, which sets its magnetic field. The two fields ie. stator and rotor field are stationary with respect to each other and develops torque. As a result of this rotor starts rotating in the direction of revolving field. It tries to catch the speed of field, but it never happens. as it leads to no induction. Hence rotor speed is always less than n_s. The difference between n_r and n_s is known as slip, ie

$$S = \frac{n_s - n_r}{n_s}$$

$\Rightarrow (n_s - n_r) = S n_s$

↑ Relative mechanical speed will induce emf in rotor ckt

$(n_s - n_r)\frac{P}{2} = S n_s * \frac{P}{2}$ will be rotor frequency ie

$$f_2 = S f_1$$

Typical value of slip may be 2-6%.

This also implies:

$$n_r = (1-S) n_s$$
$$\omega_r = (1-S) \omega_s$$

This also implies: $n_r = (1-s)\, n_s$, $\omega_r = (1-s)\, \omega_s$

Relative speeds of Rotating fields of stator and Rotor:

The combined Ampere-Turns causes rotating field in the air-gap.
The speed of the rotating field is $n_s = \frac{120 f}{P}$ rpm wrt stator.
The rotor rotates as speed n_r rpm, say for a slip 's'.
The relative speed rotating field with respect to rotor $(n_s - n_r)$.
Thus the induced voltage and current in the rotor circuit will have frequency of $f_2 = (n_s - n_r) * \frac{P}{2}$ (If n_s and n_r are in RPS)

$$= s n_s * P/2 = s * n_s * \frac{P}{2}$$

$$f_2 = s f_1$$

Now rotor circuit has current I_2. Its 3 phase equivalence will setup combined ampere-turns (ATs) and which will setup combined field of the rotor.
What is the speed of this rotor field with respect to rotor?

$$\text{Rotor field speed with respect to rotor} = \left(\frac{f_2}{P/2}\right) = \frac{s f_1}{P/2} = s n_s$$

Now rotor speed wrt stator $= n_r$
So net speed of the rotor field wrt stator $= s n_s + n_r$

$$= s n_s + (1-s) n_s$$
$$= n_s$$

By definition of slip

$$s = \frac{n_s - n_r}{n_s} \Rightarrow n_r = (1-s)\, n_s$$

$\Rightarrow$ That the rotor field and stator fields rotate at synchronous speed, n_s with respect to the rotor. And this is essential condition to develope torque in the motor.

Rotor Circuit at slip 's':

Rotor Circuit at slip 's' :

Earlier, $E_1 = 4.44 f_1 N_1 \phi_{max}$

$E_{2s} = 4.44 f_2 N_2 \phi_{max} = 4.44 s f_1 N_2 \phi_{max} = sE_2$

At stand still, $s=1$, $E_{2s} = E_2$

Rotor circuit, like secondary of will have leakage reactance x_2 and rotor resistance representing rotor copper losses as well as load, i.e.

$sE_2 = \bar{I}_2(r_2 + j\, x_{2s}) = \bar{I}_2(r_2 + j\,(2\pi f_2) L_2)$

$= \{r_2 + js(2\pi f_1) \cdot L_2\} \bar{I}_2$

$x_2 = 2\pi f_2 L_2$

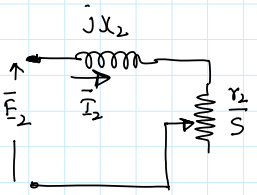
$= \omega_2 L_2$

$= (s\omega_1) L_2$

$x_{2s} = s x_2$

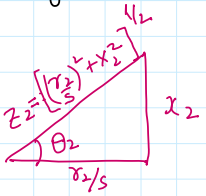
$s\bar{E}_2 = \{r_2 + j s x_2\} \bar{I}_2$

$\bar{E}_2 = \left(j x_2 + \frac{r_2}{s}\right) \bar{I}_2$



$\theta_2 = \tan^{-1}\left(\frac{x_2}{r_2/s}\right)$ = Angle between the voltage and current in the rotor.

Power factor on the rotor side, $\cos\phi_2 = \frac{r_2/s}{\sqrt{\left(\frac{r_2}{s}\right)^2 + x_2^2}}$



Power in Rotor Circuit :

The power that get transferred to the rotor circuit through air gap $= P_g = I_2^2 * \frac{r_2}{s} = E_2 I_2 \cos\phi_2 = \frac{E_2 * E_2}{\sqrt{\left(\frac{r_2}{s}\right)^2 + x_2^2}} * \frac{\frac{r_2}{s}}{\sqrt{\left(\frac{r_2}{s}\right)^2 + x_2^2}}$

$P_g = \frac{E_2^2 * r_2/s}{\{(r_2/s)^2 + x_2^2\}}$

$P_g = I_2^2 \frac{r_2}{s}$

$= I_2^2 \left\{r_2 + \frac{r_2(1-s)}{s}\right\}$

$= I_2^2 r_2 + \frac{I_2^2 * r_2(1-s)}{s}$

$= s P_g + P_g(1-s)$

$= I_2^2 r_2 + P_{em}$

= Rotor ohmic losses + Electromechanical power developed in rotor

Thus electro-mechanical torque developed in the rotor

$T_{em} = \frac{P_{em}}{\omega_r} = \frac{P_g(1-s)}{\omega_r} = \frac{P_g(1-s)}{\omega_s(1-s)} = \frac{P_g}{\omega_s} = \frac{I_2^2 r_2/s}{\omega_s}$

$T_{em} = \frac{I_2^2 r_2/s}{\omega_s} = \frac{I_2'^2 r_2'/s}{\omega_s}$

Thus, $T_{em} = \frac{P_g(1-s)}{\omega_r} = \frac{P_{em}}{\omega_r} = \frac{P_g}{\omega_s}$

$P_{em} = P_g$ – Rotor ohmic losses

$$P_{em} = P_g - \text{Rotor ohmic losses}$$

Mechanical shaft Power: $P_{sh} = P_{em} - \text{Fixed losses}$

$= P_{em} - \text{Core losses} - \text{mechanical losses}$ (friction & windage losses)

The output shaft Torque $T_{sh} = \frac{P_{sh}}{\omega_r} = \frac{P_{sh}}{\omega_s(1-s)}$
